

Sihyeon Jegal

Right to work in the UK (no sponsorship required)

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Projects

Portfolio: <https://jsihyeon.online>

Github: <https://github.com/wprkftlgus>

Summary

Full-Stack Software Engineer with hands-on experience building, deploying, and maintaining end-to-end web applications using modern front-end and back-end technologies (SaaS).

Skilled in TypeScript, Angular, React, HTML, TailwindCSS, Node.js, C#, and .NET, with experience developing RESTful APIs and working with SQL/Entity Framework Core databases.

Experienced in cloud-native deployments on AWS using EC2, ECS, and ECR, including building and deploying Docker container images, configuring SSL certificates, and monitoring services with CloudWatch.

Designed and operated a distributed architecture separating server APIs and worker services, deploying containerized APIs on AWS ECS and running background worker services on Hetzner infrastructure, coordinated via Upstash Redis for queue and task processing.

Proficient in leveraging modern AI-assisted development tools to accelerate development, improve debugging workflows, and enhance productivity across the full software development lifecycle.

Strong understanding of UI/UX principles, responsive design, and cross-browser compatibility, with experience integrating front-end applications with scalable back-end APIs and cloud-hosted databases.

Adaptable problem-solver with strong communication skills and experience working in fast-paced development environments.

Skills

Languages: JavaScript (ES6+), TypeScript, C#, HTML5, CSS3, SCSS, SQL

Frontend: React.js (Hooks, React Router, SPA), Next.js, Angular, Vite, Tailwind CSS, Bootstrap, Three.js

Backend: Node.js, Express.js, ASP.NET Core Web API, REST APIs, Middleware, CORS

Databases: MongoDB (Mongoose), MySQL, MsSQL, Entity Framework Core, AWS RDS, Upstash Redis

Cloud & DevOps: AWS (EC2, ECS, ECR, S3, RDS, IAM, CloudWatch), Azure (App Service, SQL Database), Docker (Containerization & Image Builds), Nginx, PM2, GitHub Actions (CI/CD), Certbot (HTTPS/SSL), Hetzner Cloud (Worker Service

Deployment), Redis-based Task Queues
Auth & Security: JWT, Cookies, Bcrypt, OAuth 2.0 (Google Authentication), Environment Configuration (.env)

Tools: Git, GitHub, Visual Studio Code, Microsoft Office

Practices: Responsive Web Design, Mobile-First Development, UX/UI Fundamentals, Distributed System Awareness, Unit/Integration Testing Awareness

Experience

Self-Directed Software Engineer, 01/2021 - Current

-- Leeds, UK

- Developed **8 personal and collaborative projects**, demonstrating strong skills in front-end and back-end development.
- Built a **full-stack SaaS web application**, implementing features such as authentication, database management, REST APIs, and responsive user interfaces.
- Technologies used: **React, Vite, Angular, SCSS, Tailwind CSS, Node.js, Express, C#, .NET, MySQL, MongoDB, HTML, CSS, JavaScript.**
- Designed and managed **cloud infrastructure and deployments** using **AWS (EC2, RDS, S3, IAM)** and **Azure**, ensuring secure and scalable application hosting.
- Implemented **CI/CD pipelines using GitHub Actions**, automating build, test, and deployment workflows.
- Configured **Nginx** as a reverse proxy, managed Node.js processes with **PM2**, and implemented HTTPS using **Certbot**.

- Practiced **version control with Git**, monitored production services, and maintained clear technical documentation.
- Applied **problem-solving, software design principles, and debugging skills** to deliver scalable and production-ready web applications.

Software Engineer(Frontend), 05/2023 - 12/2025

M Star Factory - Seoul, South Korea

- Developed Mezzanine Dermatology website from scratch in collaboration with another developer.
- Developed user-friendly website interfaces using HTML, CSS, and JavaScript for enhanced user experience.
- Implemented responsive web design principles to guarantee optimal viewing across various devices.
- Improved web design and usability through UX testing.

Airbnb Cohost, 03/2024 - 09/2024

Mithalia ltd - Leeds, UK

- Operated and managed a short-term rental property end-to-end, including guest communication, listings, and pricing
- Communicated with domestic and international guests to handle enquiries, bookings, and issue resolution
- Produced high-quality property photos and optimized listings to improve visibility and booking rates
- Promoted listings across multiple platforms to maximize occupancy
- Managed calendars, availability, and day-to-day operations independently

Military Security Specialist & Transport Operator, 11/2021 - 05/2023

South Korean Military - Siheung, Gyeonggi Province, South Korea

- Demonstrated discipline, reliability, and teamwork in a high-responsibility environment.
- Conducted security patrols, access control, and operational reporting.
- Operated and maintained transport vehicles for personnel, and equipment.
- Warned violators of rule infractions and escorted unauthorised people off-premises in line with company procedures.

Education

2025 University of Ulsan - South Korea

BSc (Hons) Naval Architecture & Ocean Engineering

[Degree] Graduate

- Completed a comprehensive engineering program focused on advanced mathematics, physics, mechanics, and systems analysis.
- Gained hands-on experience in structural design, CAD modelling, simulation, and computational analysis.
- Built strong expertise in fluid dynamics, thermodynamics, structural analysis, and control systems.
- Developed analytical thinking and data-driven problem-solving through engineering modelling and technical documentation.

Languages

English:	C1	Korean:	C2
Advanced		Proficient	